

Logic Detective - Developer Demo Guide & Solutions

This document contains the step-by-step logic puzzle solutions for 'Logic Detective'. Use this as a cheat sheet when demonstrating the interactive formula solver to others. By selecting the correct statements and applying the right logical laws, the game's engine will automatically simplify the equation until the target is reached.

CASE 1: THE PAWN SHOP LIFT

Variables:

P: The back door lock was picked

Q: The thief has lockpicking skills

R: Slippery Pete is our suspect

Logical Problem: $(P \rightarrow Q) \wedge (Q \rightarrow R) \wedge P \vdash R$

Target: Prove R

SOLUTION PATH 1 (Direct Modus Ponens):

1. Select $(P \rightarrow Q)$ and (P) . Click "Modus Ponens".
-> Equation simplifies, generating (Q) .
2. Select $(Q \rightarrow R)$ and (Q) . Click "Modus Ponens".
-> Equation simplifies, generating (R) .
-> Case Solved!

SOLUTION PATH 2 (Syllogism Route):

1. Select $(P \rightarrow Q)$ and $(Q \rightarrow R)$. Click "Syllogism".
-> Equation simplifies, generating $(P \rightarrow R)$.
2. Select $(P \rightarrow R)$ and (P) . Click "Modus Ponens".
-> Equation simplifies, generating (R) .
-> Case Solved!

CASE 2: THE PHANTOM BREACH

Variables:

P: The insider used a keycard

Q: The lobby alarm triggered

R: They used a remote hack

Logical Problem: $(\sim P \vee Q) \wedge (P \vee R) \wedge \sim Q \vdash R$

Target: Prove R

SOLUTION PATH 1 (Disjunctive Syllogism):

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1. Select $(\sim P \vee Q)$ and $(\sim Q)$. Click "Disjunctive Syllogism".
 - > Equation simplifies, generating $(\sim P)$.
2. Select $(P \vee R)$ and $(\sim P)$. Click "Disjunctive Syllogism".
 - > Equation simplifies, generating (R) .
 - > Case Solved!

SOLUTION PATH 2 (Equivalence & Modus Tollens):

1. Select $(\sim P \vee Q)$. Click "Equivalence".
 - > Equation simplifies, generating $(P \rightarrow Q)$.
2. Select $(P \rightarrow Q)$ and $(\sim Q)$. Click "Modus Tollens".
 - > Equation simplifies, generating $(\sim P)$.
3. Select $(P \vee R)$ and $(\sim P)$. Click "Disjunctive Syllogism".
 - > Equation simplifies, generating (R) .
 - > Case Solved!

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CASE 3: THE SILENCED WITNESS

Variables:

P: The killer fled by car

Q: The killer dropped a muddy boot

R: The killer took the subway

S: The killer took the highway

Logical Problem: $(P \wedge Q) \wedge (P \rightarrow \neg R) \wedge (\neg R \rightarrow S) \vdash S$

Target: Prove S

SOLUTION PATH 1 (Standard Deduction):

1. Select $(P \wedge Q)$. Click "Conjunctive Simp.".
-> Equation simplifies, generating (P).
2. Select $(P \rightarrow \neg R)$ and (P). Click "Modus Ponens".
-> Equation simplifies, generating $(\neg R)$.
3. Select $(\neg R \rightarrow S)$ and $(\neg R)$. Click "Modus Ponens".
-> Equation simplifies, generating (S).
-> Case Solved!

SOLUTION PATH 2 (Syllogism Route):

1. Select $(P \wedge Q)$. Click "Conjunctive Simp.".
-> Equation simplifies, generating (P).
2. Select $(P \rightarrow \neg R)$ and $(\neg R \rightarrow S)$. Click "Syllogism".
-> Equation simplifies, generating $(P \rightarrow S)$.
3. Select $(P \rightarrow S)$ and (P). Click "Modus Ponens".
-> Equation simplifies, generating (S).
-> Case Solved!

CASE 4: CHECKMATE

Variables:

Q: The blue wire is cut

R: The bomb activates

S: We hear a beep

Logical Problem: $(\neg Q \rightarrow R) \wedge (R \rightarrow S) \wedge \neg S \vdash Q$

Target: Prove Q

SOLUTION PATH 1 (Modus Tollens):

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1. Select $(R \rightarrow S)$ and $(\sim S)$. Click "Modus Tollens".
-> Equation simplifies, generating $(\sim R)$.
2. Select $(\sim Q \rightarrow R)$ and $(\sim R)$. Click "Modus Tollens".
-> Equation simplifies, generating (Q) . [Note: $\sim\sim Q$ simplifies to Q]
-> Case Solved!

SOLUTION PATH 2 (Syllogism Route):

1. Select $(\sim Q \rightarrow R)$ and $(R \rightarrow S)$. Click "Syllogism".
-> Equation simplifies, generating $(\sim Q \rightarrow S)$.
2. Select $(\sim Q \rightarrow S)$ and $(\sim S)$. Click "Modus Tollens".
-> Equation simplifies, generating (Q) .
-> Case Solved!

SOLUTION PATH 3 (Equivalence Route):

1. Select $(R \rightarrow S)$. Click "Equivalence".
-> Equation simplifies, generating $(\sim R \vee S)$.
2. Select $(\sim R \vee S)$ and $(\sim S)$. Click "Disjunctive Syllogism".
-> Equation simplifies, generating $(\sim R)$.
3. Select $(\sim Q \rightarrow R)$. Click "Equivalence".
-> Equation simplifies, generating $(Q \vee R)$.
4. Select $(Q \vee R)$ and $(\sim R)$. Click "Disjunctive Syllogism".
-> Equation simplifies, generating (Q) .
-> Case Solved!